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CSTCC 2026 OTTAWA

WHEN A FEW TREES DOMINATE THE FOREST:
OUTLIER ANALYSIS ON THERMOCHEMICAL BENCHMARKS



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Post-Doctoral Research Associate
Dalhousie Department of Chemistry
July 15, 2026

Introduction

The GMTKN55 Database

The Z-Damping Function

The XDM(Z) Dispersion Correction

What's the deal with WTMAD-2?!

Outlier Analysis on Top-Ranked GMTKN55 Functionals

Conclusions

THE GMTKN55 DATABASE

Database for general main group thermochemistry, kinetics, and noncovalent interactions

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HEAVYSB11	DARC	PX13	S22	UPU23

Five major categories: basic properties and reaction energies for small systems, reaction energies for large systems and isomerisation reactions, barrier heights, intermolecular noncovalent interactions, and intramolecular noncovalent interactions

$$\text{WTMAD-2} = \frac{1}{N_{\text{bench}}} \sum_{i=1}^{N_{\text{bench}}} \underbrace{\frac{N_{\text{syst},i}}{N_{\text{syst,mean}}}}_{\text{\# of Systems}} \cdot \underbrace{\frac{|\Delta E|_{\text{mean}}}{|\Delta E|_i}}_{\text{Mean Ref. En.}} \cdot \text{MAD}_i$$

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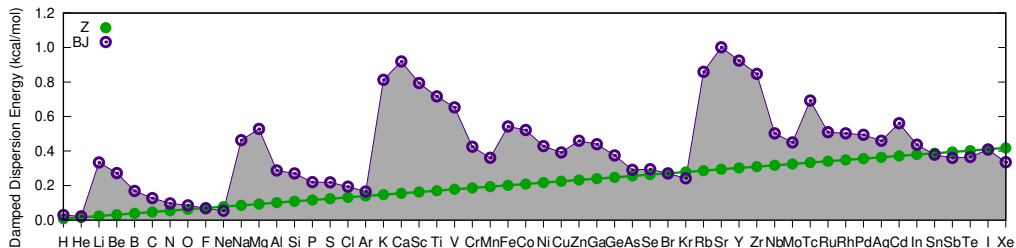
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AXEL'S ONE-PARAMETER DAMPING FUNCTION: Z DAMPING

To resolve ALK8 issues, Axel replaced the XDM dispersion correction's canonical Becke–Johnson (BJ) damping function in his DH24 double-hybrid functional.

$$E_{\text{disp}}^{\text{XDM}} = - \sum_{i < j} \sum_{n=6,8,10} f_n^{\text{damp}}(R_{ij}) \frac{C_{n,ij}}{R_{ij}^n}$$

$$f_n^{\text{BJ}}(R_{ij}) = \frac{R_{ij}^n}{R_{ij}^n + (a_1 R_{c,ij} + a_2)^n} \quad f_n^{\text{Z}}(R_{ij}) = \frac{R_{ij}^n}{R_{ij}^n + z_{\text{damp}} \frac{C_{n,ij}}{Z_i + Z_j}}$$



Becke, *J. Chem. Phys.* **160**, 204118. (2024) doi: 10.1063/5.0207682

Bryenton & Johnson, *Phys. Chem. Chem. Phys.* **28**, 11626–11638. (2026) doi: 10.1039/D6CP00403B

Rumson, Bryenton, & Johnson, *Phys. Chem. Chem. Phys.* **28**, 13543–13552. (2026) doi: 10.1039/D6CP01304J

OPEN-SOURCE IMPLEMENTATION AND TESTING OF XDM(Z)



PAPER



Cite this: DOI: 10.1039/d6cp00403b

Consistent GMTKN55 and molecular-crystal accuracy using minimally empirical DFT with XDM(Z) dispersion

Kyle R. Bryenton ^{ab} and Erin R. Johnson *^{abc}

~~OPEN-SOURCE IMPLEMENTATION AND TESTING OF XDM(Z)~~
SIDE QUEST: FIGURE OUT THE WEIRD THING HAPPENING WITH WTMAD-2



PAPER[View Article Online](#)
[View Journal](#) | [View Issue](#)



WTMAD-4: a fair weighting scheme for GMTKN55

Cite this: *Phys. Chem. Chem. Phys.*, 2026, **28**, 1463
Kyle R. Bryenton ^{ab} and Erin R. Johnson ^{★abc}

Introduction

What's the deal with WTMAD-2?!

- The issue with WTMAD-2

- Definition of WTMAD-4

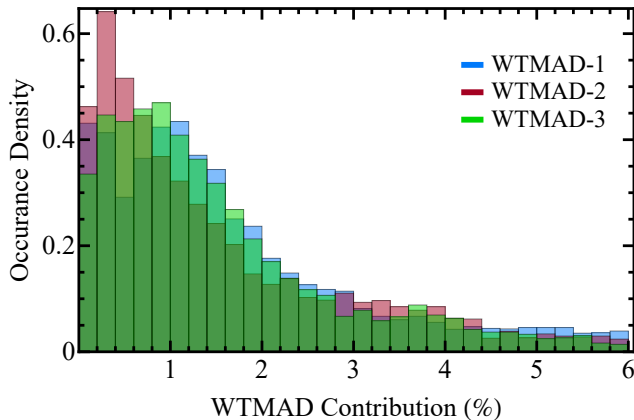
- Improvements in Statistics seen with WTMAD-4

Outlier Analysis on Top-Ranked GMTKN55 Functionals

Conclusions

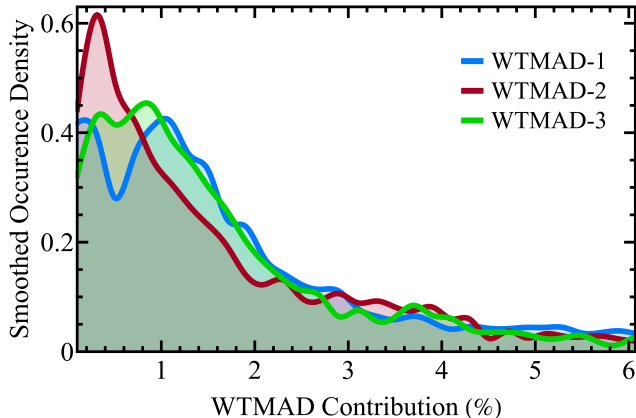
WTMAD-2 Woes

- Statistics don't form a Gaussian distribution, and have a long decaying tail
- The top 3 benchmarks contribute as much to WTMAD-2 as the bottom 36
- Benchmarks are weighed, on average, $200\times$ greater than others
- Consistent underweighting of some benchmarks (IL16: 0.06%) and overweighting of others (BH76: 9.89%)
- Poses issues for ML or parameter optimisation based on WTMAD-2



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WTMAD-2 vs WTMAD-4

- These issues are caused by WTMAD-2's dependence on the mean reaction reference energies
- Some benchmarks have $|\overline{\Delta E}|_i \approx 100$ kcal/mol with *small* average errors
- Other benchmarks have $|\overline{\Delta E}|_i \approx 1$ kcal/mol with *huge* average errors

$$\text{WTMAD-}N = \frac{1}{N_{\text{bench}}} \sum_{i=1}^{N_{\text{bench}}} w_i^{\text{WTMAD-}N} \cdot \text{MAD}_i$$

$$w_i^{\text{WTMAD-2}} = \frac{N_{\text{syst},i}}{N_{\text{syst},\text{mean}}} \frac{|\overline{\Delta E}|_{\text{mean}}}{|\overline{\Delta E}|_i}$$

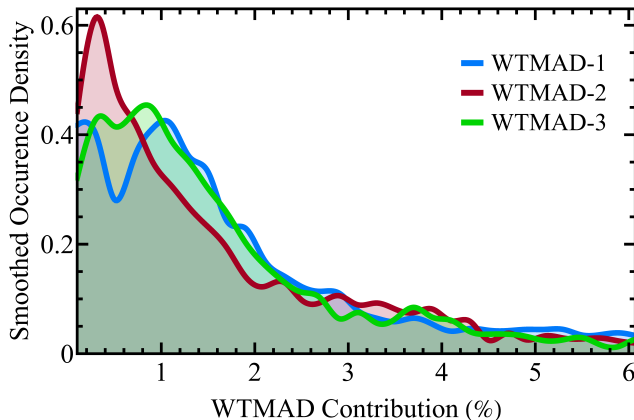
$$w_i^{\text{WTMAD-4}} = \frac{100}{N_{\text{bench}}} \frac{3.5}{\overline{\text{MAD}}_i^{10\text{-DFA}}}$$

The “10-DFA” List (incl. DFT-D3 Dispersion):

PBE0	B3LYP	B3P86	PW1PW91	HSE06
HISS	BHLYP	B3PW91	MPW1PW91	LC- ω PBE

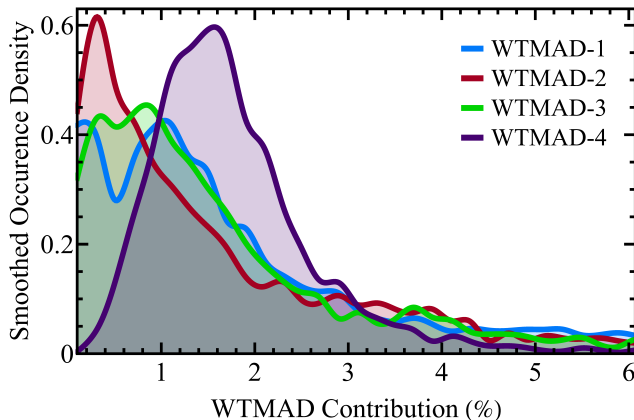
WTMAD-4 Wows

- Statistics form a more Gaussian-like distribution centred at $100\%/55 = 1.8\%$
- The top 3 benchmarks contribute as much to WTMAD-4 as the bottom 6
- Benchmarks are weighed typically within a factor of 3
- WTMAD-4 is consistent across all rungs
- Fairer treatment of benchmarks for ML uses or parameter optimisation



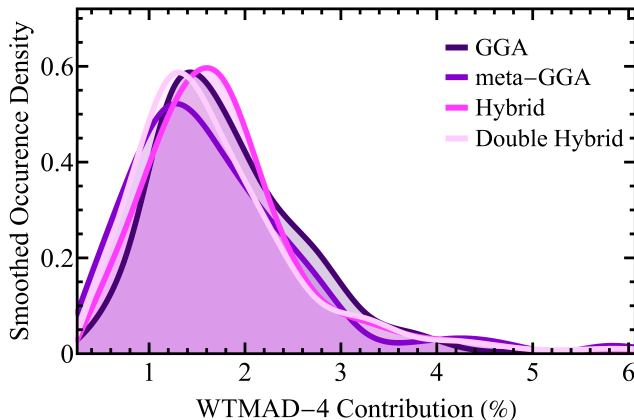
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Outlier Analysis on Top-Ranked GMTKN55 Functionals

- GGAs and meta-GGA Functionals

- Hybrid and Range-Separated Hybrid Functionals

- Double Hybrids and Machine-Learned Functionals

Conclusions

GGAs AND META-GGA FUNCTIONALS

DEFINITIONS:

$N_{r>2}$: Number of benchmarks with an error $> 2 \times$ *ratio* compared to the “10-DFA” mean

$N_{d>2}$: Number of benchmarks with an error > 2 kcal/mol *difference* compared to the “10-DFA” mean

Functional	WTMAD- <i>N</i>			$N_{r>2}$	MAX <i>r</i>	Set	$N_{d>2}$	MAX <i>d</i>	Set
	2	4	$N_{r \leq 1}$						
revPBE-D3(BJ)	8.24	7.67	16	2	2.22	BHPERI	9	9.70	SIE4x4
revPBE-XDM(Z)	8.69	7.91	19	7	2.51	SCONF	8	9.65	SIE4x4
revPBE-V	8.50	7.94	23	8	2.88	DIPCS10	11	9.36	MB16-43
OLYP-D3(BJ)	8.76	8.16	18	6	2.51	BHPERI	13	11.95	SIE4x4
B97-D3(BJ)	8.49	8.32	19	6	2.57	AL2X6	12	16.80	MB16-43
revPBE-XDM(BJ)	8.40	8.45	16	7	3.76	ACONF	11	9.79	SIE4x4
B97M-V	5.46	5.24	39	0	1.86	MB16-43	2	16.67	MB16-43
B97M-D4	5.68	5.44	37	0	1.95	MB16-43	2	18.24	MB16-43
B97M-D3(BJ)	6.44	6.41	33	3	2.81	AL2X6	3	4.97	ALK8
r^2 SCAN-D3(BJ)	7.12	6.79	25	0	1.72	WCPT18	6	4.57	SIE4x4
B97M-D3(0)	6.51	6.92	31	3	4.39	ADIM6	5	28.23	MB16-43
revTPSS-D3(BJ)	8.44	7.73	23	3	3.71	SCONF	9	17.45	MB16-43

GGAs: revPBE is still the best GGA functional for molecular chemistry
 mGGAs: Most meta-GGAs performed worse than GGAs, and often with much larger max errors

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OLYP-D3(BJ)	8.76	8.16	18	6	2.51	BHPERI	13	11.95	SIE4x4
B97-D3(BJ)	8.49	8.32	19	6	2.57	AL2X6	12	16.80	MB16-43
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HYBRID AND RANGE-SEPARATED HYBRID FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$ MAX d Set		
	2	4	$N_{r\leq 1}$						
PW6B95-D3(BJ)	5.50	5.14	41	1	2.31	IL16	0	1.76	SIE4x4
M05-2X-D3(0)	4.62	5.19	41	1	2.08	ADIM6	3	7.02	MB16-43
M06-2X-D3(0)	4.92	5.22	46	2	3.50	HEAVYSB11	1	5.83	HEAVYSB11
PW6B95-V	5.57	5.24	45	2	2.49	ADIM6	0	1.81	SIE4x4
revPBE0-D3(BJ)	5.31	5.30	43	0	1.51	W4-11	1	3.03	W4-11
revPBE0-XDM(Z)	5.71	5.43	40	0	1.48	W4-11	1	2.83	W4-11
revPBE0-XDM(BJ)	5.06	5.44	42	1	2.58	ALK8	2	7.71	ALK8
MPW1B95-D3(BJ)	5.55	5.48	42	1	2.02	ACONF	0	0.85	RSE43
B86bPBE0-XDM(BJ)	5.77	5.53	39	0	1.93	ALK8	1	4.55	ALK8
M08-HX-D3(0)	5.27	5.68	41	2	3.85	ACONF	1	2.26	C60ISO
MPW1PW91-D3(BJ)	6.33	5.82	32	0	1.31	PNICO23	0	1.04	ALK8
B86bPBE0-XDM(Z)	6.43	5.85	39	0	1.55	SCONF	0	1.68	WATER27
ω B97M-V	3.18	3.72	49	1	2.22	C60ISO	1	6.51	C60ISO
ω B97M-D4	4.02	4.34	43	1	2.24	C60ISO	2	6.60	C60ISO
ω B97X-V	3.93	4.40	50	1	2.57	C60ISO	2	13.21	MB16-43
ω B97M-D3(BJ)	3.94	4.60	44	3	2.47	C60ISO	2	7.86	C60ISO
ω B97X-D4	4.33	4.68	46	1	2.65	C60ISO	1	8.83	C60ISO

Hybrids: Significant reordering caused by the worsening of mGGA-based hybrids and the improvement of many minimally-empirical ones

RS Hybrids: The ω B97 family performs excellently, but error increased due to large MADs on C60ISO and MB16-43 benchmarks, which WTMAD-2 underweighted

HYBRID AND RANGE-SEPARATED HYBRID FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$ MAX d Set		
	2	4	$N_{r\leq 1}$						
PW6B95-D3(BJ)	5.50	5.14	41	1	2.31	IL16	0	1.76	SIE4x4
M05-2X-D3(0)	4.62	5.19	41	1	2.08	ADIM6	3	7.02	MB16-43
M06-2X-D3(0)	4.92	5.22	46	2	3.50	HEAVYSB11	1	5.83	HEAVYSB11
PW6B95-V	5.57	5.24	45	2	2.49	ADIM6	0	1.81	SIE4x4
revPBE0-D3(BJ)	5.31	5.30	43	0	1.51	W4-11	1	3.03	W4-11
revPBE0-XDM(Z)	5.71	5.43	40	0	1.48	W4-11	1	2.83	W4-11
revPBE0-XDM(BJ)	5.06	5.44	42	1	2.58	ALK8	2	7.71	ALK8
MPW1B95-D3(BJ)	5.55	5.48	42	1	2.02	ACONF	0	0.85	RSE43
B86bPBE0-XDM(BJ)	5.77	5.53	39	0	1.93	ALK8	1	4.55	ALK8
M08-HX-D3(0)	5.27	5.68	41	2	3.85	ACONF	1	2.26	C60ISO
MPW1PW91-D3(BJ)	6.33	5.82	32	0	1.31	PNICO23	0	1.04	ALK8
B86bPBE0-XDM(Z)	6.43	5.85	39	0	1.55	SCONF	0	1.68	WATER27
ω B97M-V	3.18	3.72	49	1	2.22	C60ISO	1	6.51	C60ISO
ω B97M-D4	4.02	4.34	43	1	2.24	C60ISO	2	6.60	C60ISO
ω B97X-V	3.93	4.40	50	1	2.57	C60ISO	2	13.21	MB16-43
ω B97M-D3(BJ)	3.94	4.60	44	3	2.47	C60ISO	2	7.86	C60ISO
ω B97X-D4	4.33	4.68	46	1	2.65	C60ISO	1	8.83	C60ISO

Hybrids: Significant reordering caused by the worsening of mGGA-based hybrids and the improvement of many minimally-empirical ones

RS Hybrids: The ω B97 family performs excellently, but error increased due to large MADs on C60ISO and MB16-43 benchmarks, which WTMAD-2 underweighted

HYBRID AND RANGE-SEPARATED HYBRID FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$ MAX d Set		
	2	4	$N_{r\leq 1}$						
PW6B95-D3(BJ)	5.50	5.14	41	1	2.31	IL16	0	1.76	SIE4x4
M05-2X-D3(0)	4.62	5.19	41	1	2.08	ADIM6	3	7.02	MB16-43
M06-2X-D3(0)	4.92	5.22	46	2	3.50	HEAVYSB11	1	5.83	HEAVYSB11
PW6B95-V	5.57	5.24	45	2	2.49	ADIM6	0	1.81	SIE4x4
revPBE0-D3(BJ)	5.31	5.30	43	0	1.51	W4-11	1	3.03	W4-11
revPBE0-XDM(Z)	5.71	5.43	40	0	1.48	W4-11	1	2.83	W4-11
revPBE0-XDM(BJ)	5.06	5.44	42	1	2.58	ALK8	2	7.71	ALK8
MPW1B95-D3(BJ)	5.55	5.48	42	1	2.02	ACONF	0	0.85	RSE43
B86bPBE0-XDM(BJ)	5.77	5.53	39	0	1.93	ALK8	1	4.55	ALK8
M08-HX-D3(0)	5.27	5.68	41	2	3.85	ACONF	1	2.26	C60ISO
MPW1PW91-D3(BJ)	6.33	5.82	32	0	1.31	PNICO23	0	1.04	ALK8
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ω B97M-V	3.18	3.72	49	1	2.22	C60ISO	1	6.51	C60ISO
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DOUBLE HYBRIDS AND MACHINE-LEARNED FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$	MAX d	Set
	2	4	$N_{r\leq 1}$						
DH24	1.72	2.23	52	0	1.57	C60ISO	1	3.07	C60ISO
revDH23	1.72	2.26	52	0	1.56	C60ISO	1	2.99	C60ISO
SOS-DH24	1.91	2.27	52	0	1.26	IL16	0	0.66	C60ISO
SOS-DH23	1.95	2.34	52	0	1.35	IL16	0	0.73	C60ISO
ω DOD-PBEP86-D3(BJ)	2.20	2.50	53	0	1.35	ADIM6	0	0.73	DIPCS10
ω DOD-PBEP86-D3(BJ)	2.21	2.52	53	0	1.54	ADIM6	0	0.30	DIPCS10
DM21mu	3.93	3.72	50	1	2.91	RG18	0	1.07	HEAVYSB11
DM21	3.97	3.94	48	2	2.68	RG18	1	6.01	C60ISO
DM21mc	3.96	3.98	49	1	2.12	ACONF	0	1.25	C60ISO
Skala	3.83	4.01	47	0	1.60	C60ISO	1	3.20	C60ISO
DM21m	3.89	4.04	48	1	2.98	ACONF	0	0.82	HEAVYSB11

D Hybrids: DH24 keeps the top spot, while XYG8 substantially drops from 3rd to 17th due to WTMAD-2 parametrisation

ML DFAs: Perform on par with the best range-separated hybrids, while seemingly having less extreme outliers compared to other highly-parametrised functionals of a similar rung

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Introduction

What's the deal with WTMAD-2?!

Outlier Analysis on Top-Ranked GMTKN55 Functionals

Conclusions

CONCLUDING REMARKS

1. WTMAD-2 is a misleading statistic of a functional's performance on molecular systems
2. WTMAD-4 is a much fairer representation, but...
3. Boiling 55 benchmarks down to a single number is too simplistic and more careful inspection is essential
4. Minimally-empirical functionals with good physics are still able to perform accurately and consistently

ACKNOWLEDGEMENTS



The Johnson Group:

- Prof. Erin Johnson
- Adrian Rumson
- Cameron Nickerson
- Grace Sparrow
- Laura Rizzo
- Mike Cotnam
- Miko Crouse



QUESTIONS?

WANT MY SLIDES/THESIS?



[HTTPS://GITHUB.COM/KYLEBRYENTON/SLIDES-POSTERS](https://github.com/KyleBryenton/slides-posters)

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WTMAD BoxPLOTS

